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# Personality, learning, and satisfaction in fully online academic courses

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ABSTRACT

The current study attempts to characterize the relationship between learners' personality and their satisfaction with a fully online academic course, based on previous evidence regarding the relationship between personality and self-regulated learning (SRL), and between SRL and satisfaction with online courses. Personality and satisfaction questionnaires were addressed to 72 students who were enrolled in an online academic course, in order to characterize the learners and to examine the correlation between learners' personality and their satisfaction with the fully online course. Moreover, the possibility to predict the level of the students' satisfaction according to their personality was examined as well. The findings show that openness to experience and conscientiousness significantly predicted students' satisfaction. It was found to be possible to characterize groups of online learners; and that students with similar personality traits will prefer a similar synchronous channel. A preliminary model was produced, aiming to improve learners' satisfaction from online courses.

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## 1. Introduction

*In order for educators to know*

*where their students are going,*

*they also need to know*

*where they are coming from (based on* Vygotsky, 1978*)*

Over the past two decades, higher education institutions have been offering fully online courses as part of their curriculum, along with providing credit for completion. Furthermore, the number of students not taking any online courses has continued to drop in recent years. In the same context, it is quite possible to declare that online learning is clearly mainstream (Allen, Seaman, Poulin, & Straut, 2016). Online courses are connecting elements of social networking, content from experts, and an abundance of online resources. These courses rely on the active involvement of a large number of students who independently participate according to their learning goals, prior knowledge, and skills (McAuley, Stewart, Siemens, & Cormier, 2010). However, students are dissimilar in their prior knowledge, personalities, and learning styles, all of which impact their outcomes in online courses and their success (Kauffman, 2015). Thus, notwithstanding the growth of online learning, online learning may not be suitable for every student (Bouhnik & Carmi, 2013).

Personality and learning are closely related (Eysenck, 1978). Basic personality traits may indicate ones methodical approach and the way that the individual processes information, which is considered to be a measurement tool for learning (Messick, 1984). For a long period, the Big Five personality model has been used as a predictor of performance and preferences in educational aspects; however, the majority of research in this area has focused on traditional learning methods (Keller & Karau, 2013). So far, the role that personality traits can play in this area was not taken into consideration in the research on learning and motivation to learn (Katt & Collins, 2013).

While exploring the relationship between the students' personality and fully online learning, we should not ignore the student's preferred learning style. One of the most prominent learning styles in the context of online learning is self-regulated learning (SRL). Previous studies support the claim that there is a correlation between personality traits (Big Five) and learners' SRL (Bidjerano & Dai, 2007; Ghyasi, Yazdani, & Farsani, 2013; Komarraju, Karau, Schmeck, & Avdic, 2011; Mirhashemi & Goodarzi, 2014). SRL emphasizes the active character of a learner's interactions as well as the constructive behavior with self-control in learning tasks (Martin, 2004). This type of learning style is defined as a collection of skills and learning strategies that the learner uses to handle the

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learning task efficiently and effectively (Corno, 1989, pp. 111–141). In an online environment, it is critical to understand the rule of SRL because a high level of SRL is required in online learning, even more than in face-to-face learning (Rowe & Rafferty, 2013). Previous studies found that learners who are characterized by a high tendency of SRL may find greater satisfaction with online courses (Cho & Jonassen, 2009; Nicol, 2009; Paechter, Maier, & Macher, 2010; Rowe & Rafferty, 2013).

In light of the increasing use of online courses in higher education as well as entire learning programs in the online format, and the idea that the learner is at the center of the learning process; the current study deals with the encounter between personality, learning, and satisfaction in online learning. The main aims are to characterize the online learners and to examine the correlation between learners' personality traits according to the Big Five model offered by Costa and McCrae (1985) and their satisfaction from a fully online course. Considering that the learners' inclination to SRL may indicate his/her satisfaction from the online course. Moreover, we will examine whether it is possible to predict the level of the students' satisfaction according to personality, and their preference of synchronous or asynchronous learning. Accordingly, a preliminary model will be developed.

Correspondingly, the questions in the study were: What are the personal characteristics and level of satisfaction of online learners? Is there a correlation between personality traits and satisfaction with the online course? Can students' satisfaction from online courses be predicted by their personality traits? Is it possible to characterize groups of online learners according to their personality and satisfaction with the online course? And are there differences among groups relating to synchronous or asynchronous learning?

This study sheds light on the depths of learner personality and its impact on learning process and preferences in order to understand which students will be satisfied with the online course. This study may have a significant impact on online course development and design. Furthermore, this study may impact not only on academic online courses but on other educational institutions since nowadays many of them offer training solutions and courses in the online format. Instructors and researchers can use and elaborate upon the preliminary model, developed through this study, regarding the relationship between learner personality and his/her satisfaction with fully online courses for the benefit of their research and instruction. Thus, the course could be adjusted for the learner's needs and preferences, while it is developed and taught. Adapting the online course to the learner's personality, addressing different learners, and combining various modes of learning and teaching methods will allow the learner to choose his/her preferred style. As a result, a high satisfaction could be achieved as well as low dropout rates (Park & Choi, 2009).

## 2. Literary background

### 2.1. Personality differences

In recent years, there has been a consensus that there are five dimensions that optimally feature the personality structure (Barnett, Pearson, Pearson, & Kellermanns, 2015; Caspi, Chajut, Saporta, & Beyth-Marom, 2006; Keller & Karau, 2013). These five personality dimensions are presented in the Big Five model. The purpose of these five dimensions is not to replace various names of human traits that have their distinctions and shades of meaning, but to create taxonomy, a classification system that allows us to describe all the people we know by the main dimensions that distinguish between them. This model is a method that was first developed by Fiske in 1949, but was only considered significant years later with the completion of the method development, as we know it today, by Costa & McCrae, 1985. This model is considered to be the most prominent modern psychological model (Franić, Borsboom, Dolan, & Boomsma, 2014). It was found that although the model is composed of only five traits of personality, its values scale does take into account the complexity of personality (Judge & Ilies, 2002; Otaibi & Moharib, 2012). As mentioned above, the model attributes a variety of behaviors and personality characteristics to five dimensions of character. The score that one gets on the scale enables the definition and characterization of a person:

Neuroticism – Defines the emotional stability of the individual and all aspects of anxiety, hostility, depression, self-awareness, impulsivity and emotional vulnerability. Neurotic individuals experience more negative life events because of their tendency to put themselves into situations that foster negative effects (Emmons, Diener, & Larsen, 1985).

Extraversion – This dimension refers to the individual's degree of interpersonal skills, friendliness, warmth, assertiveness, activist, thrill-seeking, and positive emotions. A person with a high score in this trait type, is characterized as more energetic, optimistic, and tends to show higher levels of commitment to social groups and activities (Watson & Clark, 1997).

Openness to experience – Describes the individual's interest in new experiences or new ideas in various fields such as fantasy, aesthetics, emotions, actions and values (McCrae, 1996). Individuals with a low score in this dimension tend to prefer the familiar and the old experiences instead of new ideas.

Agreeableness – Examines the way in which the individual interacts with his/her environment in terms of trust, direct approach, altruism, responsiveness to the needs of the environment, humility, and tenderness. Common characteristics among people with a high level of agreeableness are confidence, sympathy, willingness to help, and ability to express mercy and compassion (Patrick, 2011).

Conscientiousness – Includes the ability to control impulses, to avoid impulsive behavior, and demonstrate disciplined output of defined motives. Conscientious persons tend to maintain order, comply with obligations, reach achievements, be ambitious, have self-discipline, and be prudent in their behavior. They are organized, punctual, reliable, trustworthy, and set targets and goals for themselves (Patrick, 2011).

### 2.2. Personality and self-regulated learning

Similar to personality, learning styles represent significant differences in the individual attitudes and strategies of information processing (Snyder, 1999). In recent decades, strategies for self-regulation or self-direction have been widely studied as a component matter of learning styles (Grolnick & Raftery-Helmer, 2015; Schunk & Zimmerman, 1998). Learning styles that are accompanied by self-regulation describe a constructive and deliberate behavior during the learning process. The learner chooses learning strategies, such as setting goals, self-evaluation methods, attributing causality of success and failure, and adapts new methods for future success (Martin, 2004). Corno (1989, pp. 111–141) defined the concept of SRL as scattered skills and learning strategies, used efficiently and effectively by the learner with his/her learning task.

Studies that examined the associations between the Big Five personality traits and learning styles, characterized by SRL strategies, showed correlation among the variables (Komarraju et al., 2011). Students that tend to this learning style manage their learning process well and organize their time and their learning environment. It seems that these skills are related to some traits of personality. Most studies have indicated a correlation between conscientiousness and agreeableness dimensions with some of the SRL strategies (Bidjerano & Dai, 2007; Ghyasi et al., 2013).

Mirhashemi and Goodarzi (2014) found another correlation with the openness to experience dimension. Extroversion was associated mainly with help seeking strategies (Bidjerano & Dai, 2007; Ghyasi et al., 2013).

### 2.3. Personality, learning, and satisfaction with a fully online course

Measuring online student satisfaction has been a 'hot topic' for the academia (Horzum, 2015; Isik, 2008). It is apparent that online courses are not suitable for all learners (Bouhnik & Carmi, 2013). It seems that in order to increase the satisfaction of online learning courses, all special and possible means are required, other than the standard encouragement to achieve (Hassan, 2002). Therefore, reference to the characteristics of individual learners and their preferred learning style can be useful.

Costa and McCrae's study (1992) indicates that the five personality traits of an individual may have a strong influence on his/her learning behavior. Most online courses, given the physical distance and lack of personal acquaintance between learners who share the learning environment, require an understanding of the relationship between personality traits and online learning even more than in the traditional learning context (in the aspects of performance, behaviors, and preferences) and to assume that personality type may reflect learner preferences while internalizing information and decision-making during the online learning process (Bayne, 2004; Erdogan, Bayram, & Deniz, 2008). Barnett, Pearson, and Kellermanns (2015) found significant correlations between dimensions of the Big Five personality model and the adoption and use of technology (which is manifested in the study as a learning system). Significant positive correlations were found between technology and the conscientiousness dimension while significant negative correlations were found with the extraversion and neuroticism dimensions. In order to identify the relationship between online learning and the student's personality, the correlation between students' personality, the learning style that characterizes them, and their satisfaction with the online learning has been tested. The results showed that by referring to the personality type and the preferred learning style, an online course can be adjusted for the learner that might be more suited to his/her satisfaction. These findings have a significant impact on course development and design (Denphaisarn, 2014).

A study conducted in Macedonia set a target to produce models that attempt to predict success in learning and learners' satisfaction. The study examined the impact of personality, learning style, and satisfaction on learning outcomes. It was found that satisfaction is a factor that can predict learning outcomes, and that the learner's personality affects these outcomes and achievements (Vasileva-Stojanovska, Malinovski, Vasileva, Jovevski, & Trajkovik, 2015). In the context of satisfaction in learning and the Big Five personality model, researchers from Taiwan have found that extraversion and conscientiousness may predict satisfaction with online courses and demonstrate high motivation for learning (Shih, Chen, Chen, & Wey, 2013).

Theories and models of SRL are important for educators and those whom develop learning processes, considering the necessity to understand why some learners succeed while others have difficulties. According to the literature, there is a consensus that online learning requires higher self-regulation, and sometimes it is required more than in face-to-face learning (Rowe & Rafferty, 2013). SRL has an impact on the nature of the behavior of learners in an online environment, both in terms of achievements and interactions with colleagues (Lin, Huang, & Chuang, 2015). In studies that examined technologically integrated courses that allowed students to choose different strategies of self-regulation, a high level of satisfaction and achievement was found. Online learning extends the content of learning interactions and allows for the use of additional interactions in the online world, which are not necessarily human interactions (Cho & Jonassen, 2009; Nicol, 2009).

The synchronous learning platform is one of the online learning options that is possible in the present era, as a result of technological progress. In their study, Blau and Barak (2012) found that learner personality has an influence on learner involvement in synchronous learning platforms; beyond the communication device that is used and the discussion subject, it was found that personality traits have influence on synchronous online behavior. According to the results of their study, the researchers claimed that the learning environment can be adjusted for learners with different personality traits.

The presented literary background leads to the claim that adequate knowledge about the direct relationship between personality traits and satisfaction with online courses is not provided (Bishop-Clark, Dietz-Uhler, & Fisher, 2007; Bolliger & Erichsen, 2012; Keller & Karau, 2013; Shih et al., 2013). Even though there is previous evidence regarding the relationship between personality and SRL (Bidjerano & Dai, 2007; Ghyasi et al., 2013; Komarraju et al., 2011; Mirhashemi & Goodarzi, 2014), as well as between SRL and satisfaction with online courses (Cho & Jonassen, 2009; Nicol, 2009; Paechter et al., 2010; Rowe & Rafferty, 2013), the relationship between the personality traits and satisfaction with online courses has not been studied enough, and therefore it is uncertain (Keller & Karau, 2013; Shih et al., 2013; Tlili, Essalmi, Jemni, & Chen, 2016). Identifying personal characteristics that contribute to learner satisfaction may help to increase success in online courses. Moreover, this awareness of learners' characteristics may help in designing high-quality online courses that meet the needs of learners and improve the level of achievement and satisfaction from the course (Kauffman, 2015; Schrum, 1995).

## 3. The research

In light of the growing trends of fully online courses offered by higher education, along with the adoption of the learner-centric approach, this study focuses on shedding light on the influence of learners' personalities on their satisfaction with online courses. In this research, the standard definition of online courses was used, in which at least 80% of the instruction/materials are presented online (Allen et al., 2016) with no classroom attendance required (Keller & Karau, 2013).

### 3.1. Research aims and questions

The current study focuses on the encounter between students' personality, online learning, and satisfaction with fully online courses. The study aims to examine and characterize online learners and to test the correlation between students' personality traits, according to the Big Five model offered by Costa and McCrae (1985), and their satisfaction with the fully online course, taking into account that SRL might affect higher satisfaction with the online course. Thus, three aspects were explored through the research questions:

a) Personality characteristics and learners' satisfaction from the online course
   - Q1: What are the personal characteristics of the online course students?
   - Q2: What is the level of satisfaction of the online students?
b) The relationship between personality traits and satisfaction from online courses

Q3: Is there a correlation between personality traits and satisfaction with online courses?

Q4: Can students' satisfaction from online courses be predicted by their personality traits?

c) Online learners' characterization by their personality and their satisfaction with the online course

Q5: Is it possible to characterize groups of online learners according to their personality and satisfaction with the online course?

Q6: Are there differences among groups relating to synchronous or asynchronous learning?

### 3.2. Research hypotheses

This study hypothesized that correlations between personality traits and learners' satisfaction with online courses would be found. This hypothesis is based on previous studies which support the claim that there is a correlation between personality traits (Big Five) and learners' SRL (Bidjerano & Dai, 2007; Ghyasi et al., 2013; Komarraju et al., 2011; Mirhashemi & Goodarzi, 2014) as well as the claim that learners who are characterized by a high tendency of self-regulated learning may find greater satisfaction with online courses (Cho & Jonassen, 2009; Nicol, 2009; Paechter et al., 2010; Rowe & Rafferty, 2013). Fig. 1 presents the research framework regarding the relationships between personality traits and learners' satisfaction based on the presented literature.

More specifically, three dimensions of personality were found to be related to SRL and enable the prediction of students' satisfaction with online courses: conscientiousness, agreeableness (Bidjerano & Dai, 2007; Mirhashemi & Goodarzi, 2014), and openness to experience (Komarraju et al., 2011; Mirhashemi & Goodarzi, 2014). In addition, Ghyasi et al. (2013) found correlation between conscientiousness and SRL, but not between agreeableness and SRL; while Komarraju et al. (2011) found a general correlation between conscientiousness and learning style. However, there was insufficient research to support the correlation between extraversion, neuroticism, and SRL. In regard to neuroticism, Komarraju et al. (2011) found negative correlation with SRL.

Thus, hypotheses regarding each trait were examined concerning satisfaction, believing that students with a high level of conscientiousness, openness to experience, and agreeableness would be more satisfied with the online course, compared to others.

**H1**. High significant correlations will be found between satisfaction with the online course and conscientiousness, openness to experience, and agreeableness.

**H2**. No significant correlations will be found between satisfaction from the online course, extraversion, and neuroticism.

**H3**. Learner personality traits: conscientiousness, openness to experience, and agreeableness, will have significant effect on learner's satisfaction.

In addition, the current study aims to classify groups of students based on their personality traits and their satisfaction with the online course. The classification will enable differentiation between groups of students, while still showing similarities within each group according to the study topics (personality traits and satisfaction with the online course). Thus,

**H4**. Groups of students with similar levels of personality traits will show different levels of satisfaction with the online course.

Blau and Barak (2012) claimed that there is a correlation between personality and learners' usage and behavior in synchronous channels, therefore,

**H5**. Learners with similar personality traits will have a similar preference for synchronized learning.

## 4. Methodology

In order to answer the research questions, an anonymous questionnaire was distributed to 72 students who participated in a fully online course at a large, accredited university. The course was delivered at the Teacher Education Unit of the university. All students had a BA degree in their discipline. The average age of the

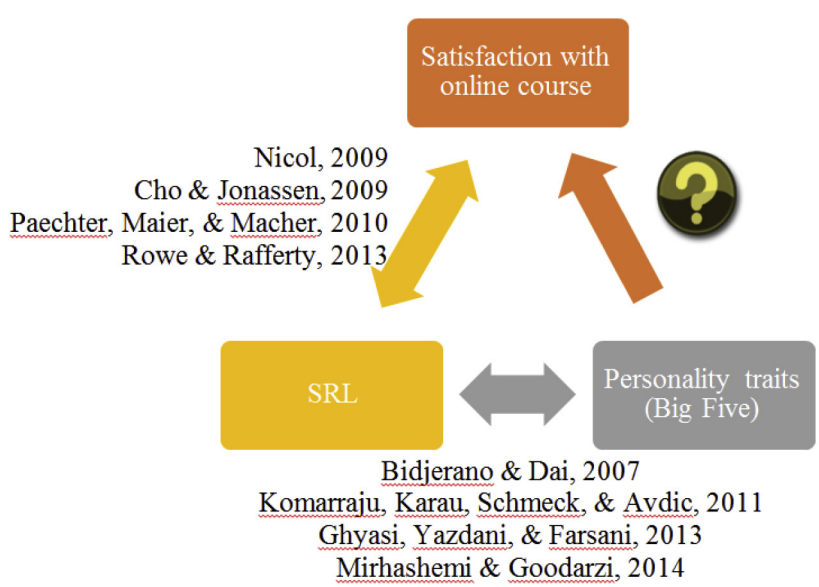


**Fig. 1.** Relationships between personality traits, SRL, and learners' satisfaction.

students was 30 years old. The majority of the students were between the ages of 21–31 (69%); 25% of them were between the ages of 32–42, and the rest (6%) were between the ages of 43–60. Notably, 63% of the students were female while 37% were male.

The questionnaire comprised two sub-questionnaires: The first, BFI (Big Five Inventory) according to the version of John and Srivastava (1999, pp. 102–138), a shortened version of the 44 statements from the original Big Five questionnaire. The respondents were asked to rate their agreement with 44 statements that characterize their personality in ordinal scale from 1 to 5 (1 meaning strongly disagree and 5 meaning strongly agree). The questionnaire examined the five traits, as described in The Big Five Model. Each trait was examined with various statements as follows: extraversion - 8 statements ($\alpha = 0.81 > 0.6$); neuroticism - 8 statements ($\alpha = 0.89 > 0.6$); agreeableness - 9 statements ($\alpha = 0.84 > 0.6$); conscientiousness - 9 statements ($\alpha = 0.78 > 0.6$); and openness to experience - 10 statements ($\alpha = 0.73 > 0.6$). An average score for each dimension was calculated according to the scores of each participant in the relevant statement. Some statements in the questionnaire underwent a reversal. The second sub-questionnaire was developed especially for this study and included 10 statements for examining the students' satisfaction with the online course. The respondents were asked to rate their agreement with the presented statements in ordinal scale from 1 to 5 (1 meaning strongly disagree and 5 meaning strongly agree). This sub-questionnaire focused on the communication quality in the online course, the learner's desire for synchronous/asynchronous learning, their motivation to learn, and the interest that the course aroused in them. Furthermore, the participants were asked to assess their degree of overall satisfaction with the course. Reliability analysis was conducted to examine the internal consistency of the satisfaction sub-questionnaire. The analysis revealed high reliability for this sub-questionnaire (N = 10., $\alpha = 0.855 > 0.6$).

After receiving replies from the students, seven scores were calculated for each student: one for each of the Big Five dimensions, one for satisfaction with the online course, and one for synchronous learning preference. Following that, statistical analyses were performed. Regarding Q1and Q2, theoretical statistics for the personality traits of the students and their level of satisfaction with the online course were conducted. Spearman correlation analyses were conducted between personality traits and satisfaction with the online course in order to answer Q3, and regression analysis was performed in order to predict students' satisfaction with the online course according to their personality traits (Q4). In addition, cluster analysis was conducted in order to characterize groups of learners according to their personality and satisfaction with the course. Following that, an ANOVA analysis was performed to examine the significance of the differences among the yielded groups (Q5). Finally, ANOVA analysis performed to examine the differences among groups regarding synchronous learning preference (Q6).

## 5. Findings

### 5.1. Personality traits and satisfaction of the online course students - descriptive statistics

In order to answer the first question (Q1) students' personality characteristics were explored using the Big Five model offered by Costa and McCrae (1985). It was found that conscientiousness was the highest scoring trait characterizing the students (Mean = 4.01). Agreeableness also characterized the students with a high score (Mean = 3.92). However, neuroticism was not found to have a high score (Mean = 2.38); in fact, it was found to score the lowest (MIN = 1.22). The maximum scores were found in both agreeableness and conscientiousness (5.00). The widest range of scores and differences within a group were found in the extroversion trait (SD 0.729), whereas openness to experience had the smallest range of scores (0.503). In addition, the results showed that the students' data in this study was normally distributed based on the degrees of skewness and kurtosis. Skewness values were less than the absolute value 0.5 and kurtosis values were less than the absolute value 1 (see Table 1).

**Table 1**
Descriptive statistics of each trait.

| | Median | Mode | M | SD | Skewness | Std. Error of skewness | Kurtosis | Std. Error of Kurtosis | Minimum | Maximum |
|---|---|---|---|---|---|---|---|---|---|---|
| Openness to experience | 3.67 | 3 | 3.66 | 0.503 | 0.088 | 0.283 | −0.422 | 0.559 | 2.5 | 4.8 |
| Neuroticism | 2.44 | 2 | 2.38 | 0.63 | 0.209 | 0.283 | −0.269 | 0.559 | 1.22 | 4 |
| Conscientiousness | 4 | 4 | 4.01 | 0.545 | −0.204 | 0.283 | −0.356 | 0.559 | 2.63 | 5 |
| Agreeableness | 4 | 4 | 3.92 | 0.551 | −0.526 | 0.283 | 0.26 | 0.559 | 2.44 | 5 |
| Extraversion | 3.56 | 4 | 3.47 | 0.729 | −0.107 | 0.283 | −0.586 | 0.559 | 1.63 | 4.88 |

Distributions of students according to each trait are shown in Figs. 2–6.

Students' distribution divided by low, medium, and high scoring personality traits, was also tested (Fig. 7). It was found that neuroticism had the most students who received a low score (59 of 72 students) while conscientiousness had the most students who received a high score (42 students). The trait in which most students received a medium score was openness to experience (45 students). In addition, it can be seen prominently that most

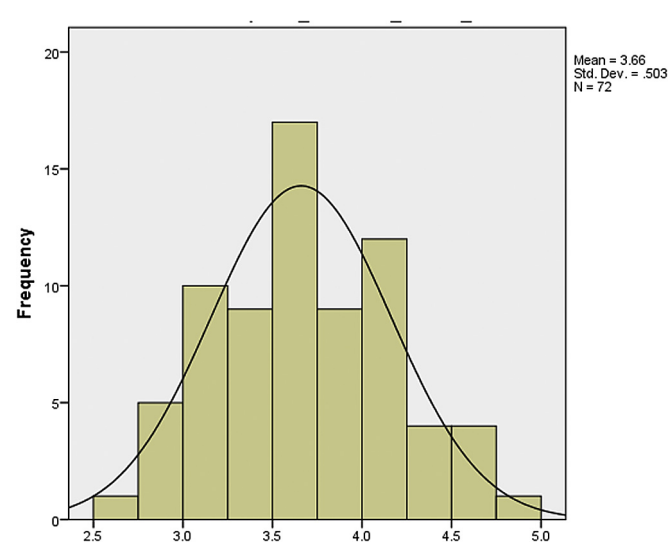


**Fig. 2.** Openness to experience trait distribution.

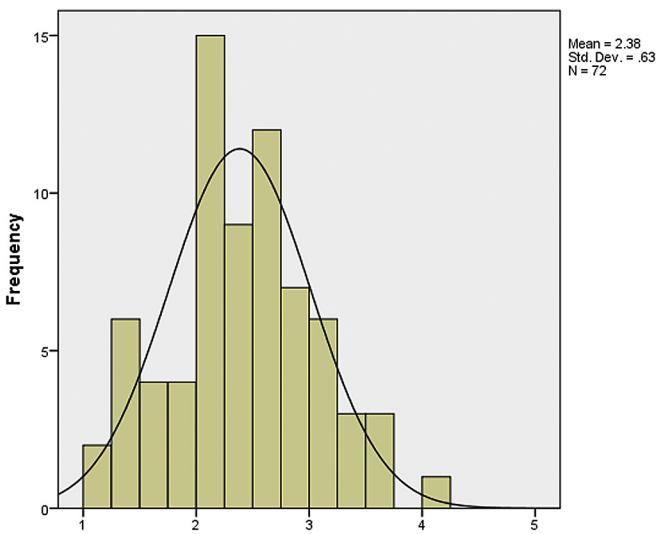


**Fig. 3.** Neuroticism trait distribution.

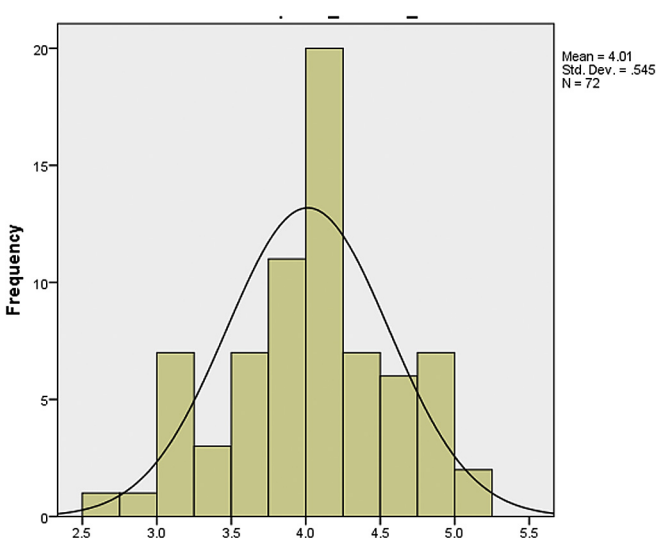


**Fig. 4.** Conscientiousness trait distribution.

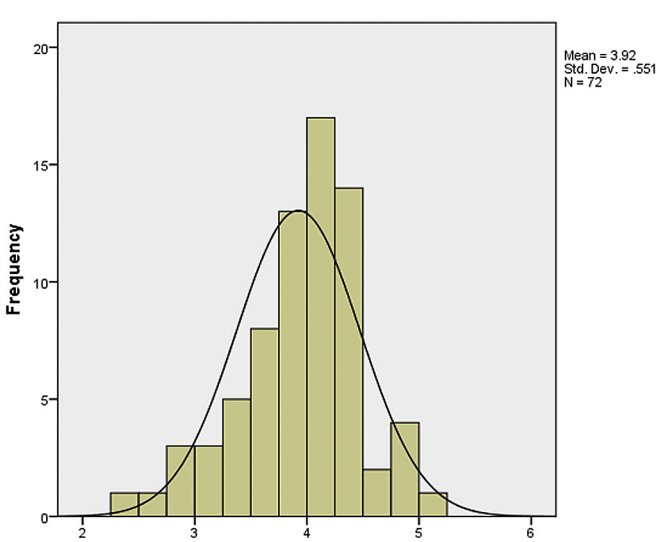


**Fig. 5.** Agreeableness trait distribution.

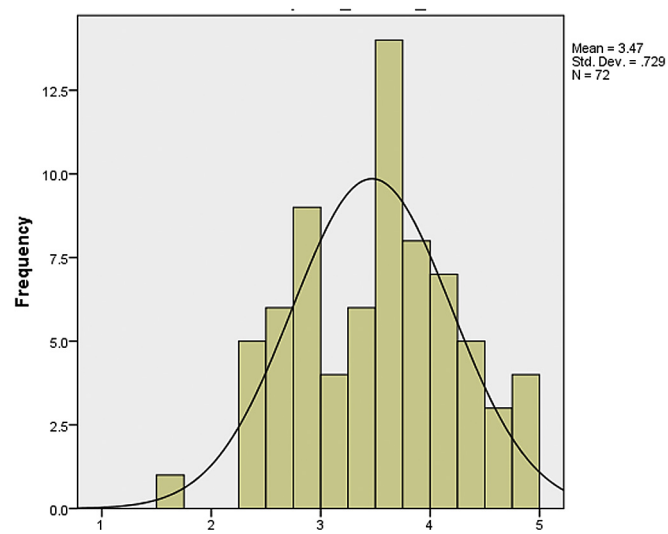


**Fig. 6.** Extraversion trait distribution.

students did not have a tendency to neuroticism while they had a high level of conscientiousness and agreeableness (Fig. 7).

Regarding the second question (Q2) which deals with students' satisfaction with the online course (Table 2), it was found that the average satisfaction level was moderate (M = 3.35, SD = 1.199). In this case (as opposed to personality traits), students' distribution was not normal, although it tended to have symmetry (kurtosis>1, skewness<0.5). Students' satisfaction distribution showed that almost the same number of students received a low or high satisfaction score (28 low satisfaction and 29 high satisfaction). Only 15 students received a moderate score.

The distribution of the students' satisfaction with the online course is presented in Fig. 8.

### *5.2. Correlations between personality traits and students' satisfaction with the online course*

In order to test the first and the second hypotheses (Q3), Spearman analysis was conducted to examine the correlation between the personality traits and the level of satisfaction with the online course. The first hypothesis (H1) was that high significant correlations will be found between satisfaction with the online course and conscientiousness, openness to experience, and agreeableness. The analysis revealed (Table 3) a positive, moderate to high significant correlation between conscientiousness and satisfaction ($r = 0.39$, $p < 0.01$) as well as between openness to experience and satisfaction ($r = 0.38$, $p < 0.01$). Learners with high scores in conscientiousness and openness to experience were more satisfied with online learning than those with lower scores in these areas. However, regarding agreeableness, no correlation was found with satisfaction ($r = 0.10$, $p > 0.05$). Hence, H1 was partially supported by this research. This finding was consistent with expectations only in regard to the relationship between the conscientiousness and openness to experience traits, and satisfaction. The more learners are defined as conscientious and with openness to experience, the greater their satisfaction with online courses will be, and vice versa. However, regarding the relationship between agreeableness and satisfaction, the finding was not consistent with expectations.

The second hypothesis (H2) was that no significant correlations will be found between satisfaction from the online course, extraversion, and neuroticism. Indeed, no correlations were found

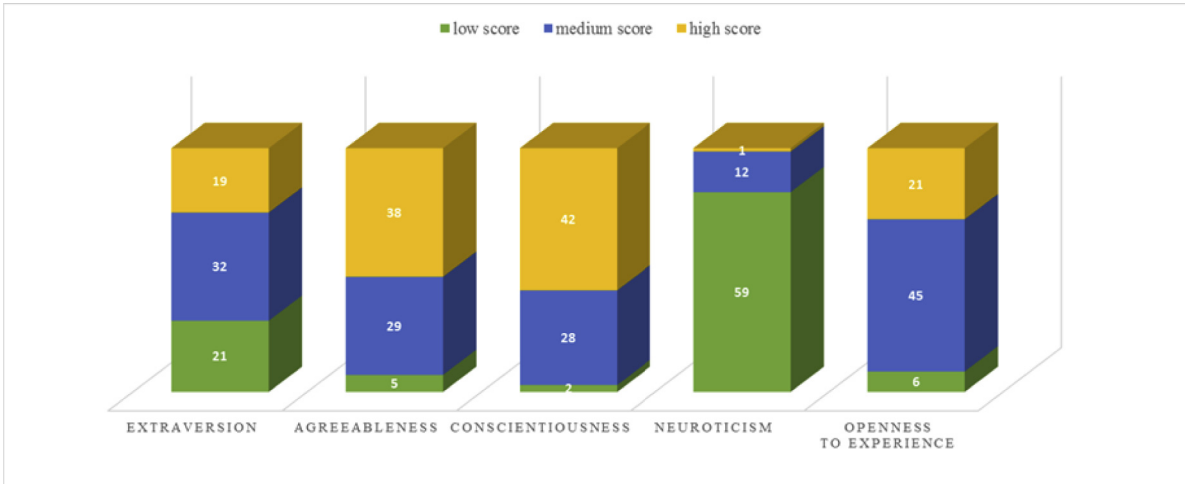


**Fig. 7.** Students' distribution by personality trait.

**Table 2**
Descriptive statistics of students' satisfaction.

| | Median | Mode | M | SD | Skewness | Std. error of skewness | Kurtosis | Std. Error of Kurtosis | Minimum | Maximum |
|---|---|---|---|---|---|---|---|---|---|---|
| Satisfaction | 3.63 | 5 | 3.35 | 1.199 | -0.368 | 0.283 | −1.188 | 0.559 | 1 | 5 |

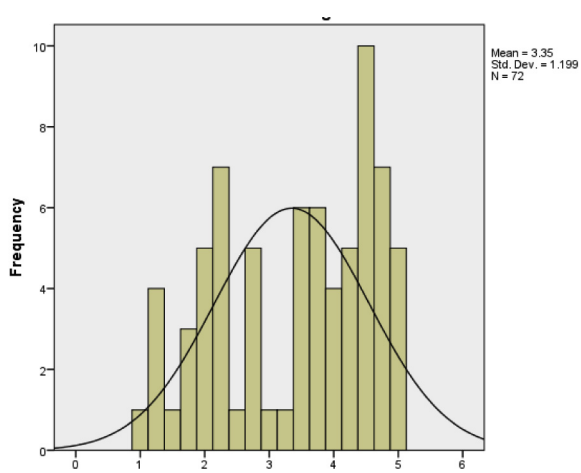


**Fig. 8.** Distribution of the students' satisfaction.

between satisfaction and extraversion ($r = 0.025$, $p > 0.05$) or satisfaction and neuroticism ($r = 0.04$, $p > 0.05$). Hence, the second hypothesis (H2) was supported. This finding was consistent with expectations.

Table 4 presents the correlation values between personality traits and each of the questionnaire statements regarding satisfaction with the online course.

**Table 3**
Correlations between personality traits and satisfaction.

| Personality traits | Student satisfaction score with the online course |
|---|---|
| Openness to experience score | **0.376**** |
| Neuroticism score | 0.041 |
| Conscientiousness score | **0.390**** |
| Agreeableness score | 0.099 |
| Extraversion score | 0.025 |

**p < 0.01.

In order to test the third hypothesis (H3), that the learner personality traits of conscientiousness, openness to experience, and agreeableness will have significant effect on learner's satisfaction (Q4), linear regression analysis (multiple regression) was performed and the five trait variables were added to the model. A statistically significant model was obtained, explaining 30% of the variance of students' satisfaction ($F$ (5,66) = 5.71, $p < 0.001$). The relative impact of only two traits was found to be significant: openness to experience and conscientiousness, whereas the relative impact of agreeableness was not found. Hence, H3 was partially supported. The relative impact of all traits is shown in Table 5. The traits that are highlighted in the table significantly predicted students' satisfaction.

### *5.3. Characterization of groups of students according to their personality and their satisfaction with the online course*

An attempt to characterize groups of students based on their personality traits and their satisfaction with the online course was made in this study using two step cluster analysis. This was done to test the fourth hypothesis (H4) that groups of students with similar levels of personality traits will show different levels of satisfaction with the online course (Q5). The variables included the five personality trait scores and students' satisfaction scores. Three clusters resulted with good quality and a relatively high silhouette measure of cohesion and separation (average silhouette = 0.5). Table 6 presents the generated groups with relation to the students' personality trait and satisfaction scores.

As presented in Table 6, different levels of satisfaction with the online course were found in each group. Group 1 was characterized with a high level of satisfaction. This group consisted of students with high scores in extraversion, agreeableness, conscientiousness, and openness to experience, but not in neuroticism. Furthermore, their neuroticism scores were the lowest of all the groups. Group 2 was characterized with a low level of satisfaction, consisted of students with high neuroticism scores in relation to the other

**Table 4**
Correlation between personality traits and satisfaction statements.

| | Openness to experience score | Neuroticism score | Conscientiousness score | Agreeableness score | Extraversion score |
|---|---|---|---|---|---|
| Course staff provided responses to my requests. | **0.344**** | -0.017 | 0.033 | 0.047 | 0.067 |
| I would like face-to-face meetings with faculty to be integrated. | -0.060 | 0.254* | -0.116 | -0.069 | 0.164 |
| I would like synchronous online meetings with faculty to be integrated. | 0.242* | 0.218 | -0.039 | 0.008 | 0.058 |
| Communication with other students was satisfying. | 0.269* | 0.218 | 0.209 | 0.120 | -0.044 |
| I felt highly motivated while learning online. | 0.282* | 0.019 | **0.408**** | 0.044 | -0.074 |
| The course helped me greatly to understand the learning themes. | **0.305**** | 0.006 | **0.379**** | 0.074 | 0.032 |
| The course was organized and clearly delivered. | 0.226 | -0.126 | **0.384**** | 0.029 | -0.054 |
| The course was interesting. | **0.310**** | 0.025 | **0.417**** | 0.116 | 0.037 |
| The course triggered intellectual curiosity. | 0.288* | 0.019 | **0.400**** | 0.088 | 0.085 |
| I am generally satisfied with the course. | **0.317**** | 0.081 | **0.322**** | 0.101 | -0.065 |

*$p < 0.05$; **$p < 0.01$.

**Table 5**
Regression results for predicting students' satisfaction (N = 71).

| Traits | β | T | P |
|---|---|---|---|
| (Constant) | | −1.792 | 0.078 |
| Openness to experience score | 0.416 | 3.658 | **0.001** |
| Conscientiousness score | 0.451 | 4.061 | **0.000** |
| Agreeableness score | -0.067 | -0.597 | 0.553 |
| Extraversion score | -0.212 | −1.868 | 0.066 |
| Neuroticism score | 0.074 | 0.683 | 0.497 |

**Table 6**
Cluster descriptions [the colors represent the input (predictor) importance].

| | Group 1 | Group 2 | Group 3 |
|---|---|---|---|
| Size (N) | 19 (26.4) | 23 (31.9%) | 30 (41.7%) |
| Openness to experience | 3.92 | 3.15 | 3.89 |
| Conscientiousness | 4.58 | 3.94 | 3.71 |
| Satisfaction | 4.35 | 2.33 | 3.50 |
| Extraversion | 4.01 | 3.22 | 3.32 |
| Agreeableness | 4.28 | 3.76 | 3.82 |
| Neuroticism | 2.19 | 2.47 | 2.44 |

groups, and had the lowest scores in openness to experience, agreeableness, and extraversion. Group 3 was characterized with a moderate level of satisfaction and consisted of students with the lowest scores in conscientiousness. Fig. 9 graphically presents the cluster analysis findings and prominently shows where the members of each group are located in terms of personality traits and level of satisfaction with the online course in relation to the population being tested.

In order to show the significant differences among the three clusters regarding their level of satisfaction with the online course, a one-way ANOVA analysis was performed.

The ANOVA yielded significant differences among the three clusters [$F(72) = 25.531$, $P < 0.001$]. In addition, significant differences in satisfaction among the groups were found regarding synchronous learning in general, whether it was online or face-to-face [$F(72) = 3.273$, $P < 0.05$] as well as, in particular, to online synchronous learning [$F(72) = 6.838$, $P < 0.001$].

In light of the presented findings, it can be claimed that the hypotheses, H4, that groups of students with similar levels of personality traits will show different levels of satisfaction with the

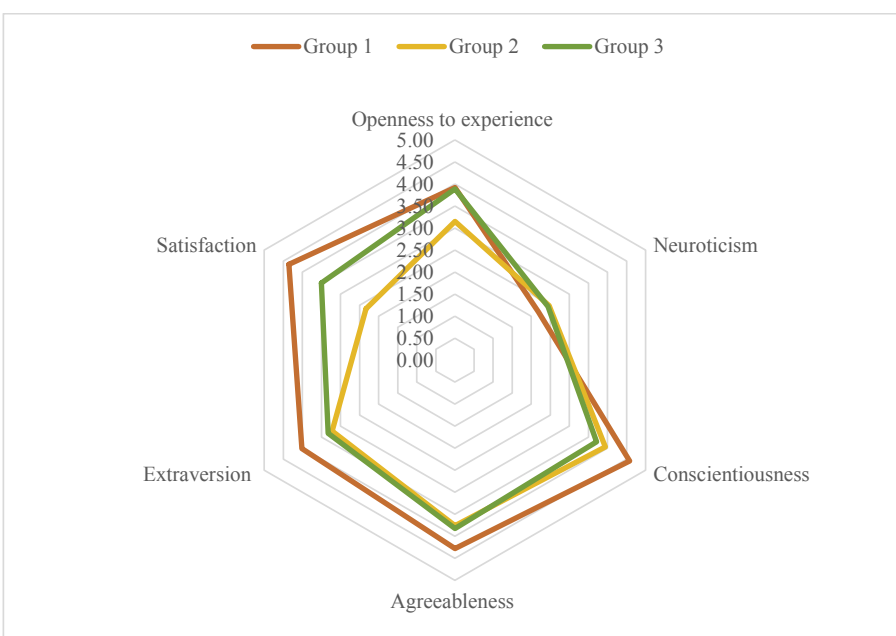


**Fig. 9.** Graphic view of cluster analysis findings.

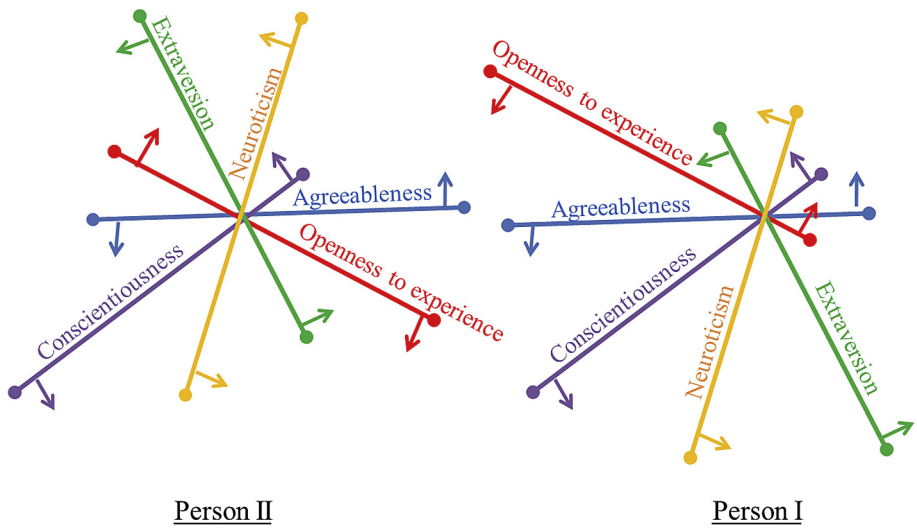


**Fig. 10.** Graphic illustrations of two examples of personality structure according to the Big Five model.

online course, and H5, that learners with similar personality traits will have a similar preference for synchronized learning, were supported. As expected, groups of students with similar levels of traits, indeed, showed different satisfaction levels with the online course. Furthermore, students with similar levels of traits had a similar preference for synchronized learning.

## 6. Discussion

### *6.1. Personality traits and their relation to satisfaction with an online course*

Online courses are a growing trend in all institutions of education, especially in higher education (Allen et al., 2016). However, online learning may not fit every student (Bouhnik & Carmi, 2013; Kauffman, 2015). Considerable research has been conducted on fully online courses but not enough has been focused on the personality traits of the learners in that context (Orvis, Brusso, Wasserman, & Fisher, 2010; Tlili et al., 2016). Knowing the students' characteristic traits may assist in meeting their learning needs and raise satisfaction. As aforementioned, according to the Big Five model, each person has all five traits in different levels of tendency (Patrick, 2011). The Big Five model was chosen for being a famous hierarchical model with five bipolar dimensions (traits), which allow for personal examination in a broad and abstract manner (Judge & Ilies, 2002; Otaibi & Moharib, 2012). The personality structure of each person places him/her on a different point along the dimension and thereby reflects interpersonal differences. Fig. 10 graphically illustrates personality examples of two different people.

The research hypotheses were determined based on relationships which were previous found between personality traits and SRL. It is important to note that the students in this research did not complete a questionnaire examining SRL because of the assumption that the relationship between personality traits and SRL had already been tested and measured, as described earlier in the literature review. In addition, the relationship between SRL and learners' satisfaction with the online course was not tested since it had already been proven (Cho & Jonassen, 2009; Paechter et al., 2010). Rowe and Rafferty (2013) claimed that online learning requires SRL at a high level. Furthermore, Nicol (2009) revealed that students showed greater independence in online learning, and independent learning is a central feature of SRL style. Joo, Joung, and Kim (2014) found an indirect relationship between these variables; however, Kuo, Walker, Belland, and Schroder (2013) failed at doing that. Still, a scientific basis indicating that there is a relationship between the variables does exist.

The current study findings show that students who tend to high openness to experience and conscientiousness are more satisfied with online courses. The relationship between these two traits and the students' satisfaction is consistent with previous studies that found a relationship between openness to experience, conscientiousness, agreeableness, and the tendency to use strategies of SRL (Bidjerano & Dai, 2007; Ghyasi et al., 2013; Komarraju et al., 2011; Mirhashemi & Goodarzi, 2014). However, these previous studies found a relationship between agreeableness and SRL, while the current study did not find a correlation between students' satisfaction and agreeableness.

In the absence of sufficient evidence in previous research for the relationship between extraversion and neuroticism, and SRL or satisfaction from online courses, the current research hypothesized that no correlation would be found between extraversion and neuroticism, and satisfaction with the online course. Indeed, no correlation was found between those traits and satisfaction; however, it is important to note that one of the three groups of students that were yielded based on their personality traits and their satisfaction with the online course was characterized by the lowest level of satisfaction as well as a high level of neuroticism. In addition, the group that was characterized with a high level of satisfaction consisted of students with high scores in extraversion, agreeableness, conscientiousness, and openness to experience, but not in neuroticism (which was the lowest of all the groups). Moreover, groups of students with similar levels of traits have a similar preference for synchronized learning, in general, whether it is online or face-to-face. This supports Blau and Barak's findings (2012), which proved a correlation between personality characteristics (although not by the Big Five model) and the learners' synchronous behavior. Thus, the yielded groups reflect the differences between the groups of students; however, within each group they reflect similarities.

Showing that there are relationships between personality traits and satisfaction from online courses, and moreover that learners with similar personality traits will probably have similar online learning preferences, may shed light on the online learners. This

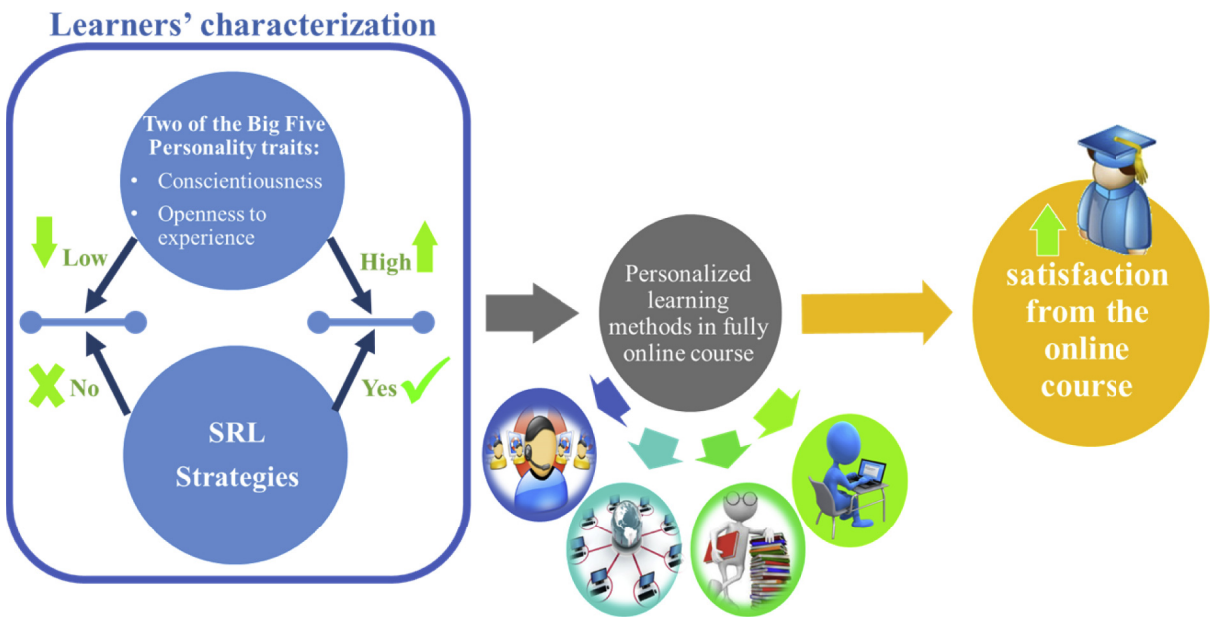


**Fig. 11.** A preliminary model for fostering learner satisfaction.

gives a basis for new, online, pedagogical approaches in addition to improving the development processes of online courses, under the assumption that it is possible to customize courses to various types of learners who have different personality traits and different learning styles. This assumption is significant for educational institutions and for the training world where online courses are integrated for various aims: improving learning quality and efficiency, as well as learners' satisfaction (Appana, 2008; Bartley & Golek, 2004).

The current research has focused on personality traits, with the aim of placing them in the center of the research debate regarding online course development and instruction. The results indicate that the tendency of some students to have low conscientiousness and/or openness to experience affects their satisfaction with online courses. This is important to take into consideration at the development stage as well as during the instruction and learning processes, and to try to customize the course format to students with various traits. In subsequent research, we aim to produce significant insight into academic courses and development in the training world regarding the possibility to produce online learning platforms that suit a wide variety of learner types, focusing on the learners' trait levels (Judge & Ilies, 2002; Otaibi & Moharib, 2012). It is also important to note the research limitation of the relatively small sample size. In order to develop an instructional model that fits the variety of student personalities, further research should be conducted with a larger population as well as a range of learning styles.

### 6.2. A preliminary model for fostering learner satisfaction

In light of the findings of this research and the claim that fully online courses do not fit all learner types (Bouhnik & Carmi, 2013) we want to suggest a preliminary model for fostering students' satisfaction that can be explored in future research (Fig. 11). This preliminary model is based on the literature which deals with active and independent learners (Zimmerman, 2008), learning style, SRL, and the mechanisms used to evaluate those who tend to self-guidance (Boekaerts, 1999). The aim of the model is to describe the process of online course development with an emphasis on adaptation to the learners' needs, preferences, and learning style (McManus, 2000; Tallent-Runnels et al., 2006). During online course development, adaptation should be considered in regard to learner personality and preferred learning style, under the assumption that the online learning environment may be designed with multiple methods that fit learners with different personality traits (Blau & Barak, 2012).

In order to develop an appropriate online course for those who tend to conscientiousness and openness to experience, it is important to incorporate educational methods and teaching aids which allow for independence and control over the pace of learning. These learners do not require provisions or guidance from others, thus freer learning is possible (McManus, 2000). The same applies to the assessment mechanisms; these learners do not need external regulation and there is no need for increased human or computerized feedback and coaching (Cho & Jonassen, 2009). Conversely, additional educational aids and material need to be produced and integrated in the course for the students who have low conscientiousness and openness to experience. These learners will benefit from a more supportive learning environment, less independent demand, and accompanied feedback and external support (not necessarily human). With this model it will be possible to develop a versatile course in which each learner may find a satisfying alternative and preferred learning style to support them in completing it successfully; based on the supported hypothesis that adapting online instruction to learner personality will raise the level of student satisfaction.

The current research findings and insights that led to the proposed model are consistent with the Denphaisarn study (2014), which showed that by addressing personality type and preferred learning style, an online course can be adjusted to the learner and raise satisfaction. Investment in the course development phase and integration of various modes of teaching and teaching methods for different personality traits and learning styles may affect learner satisfaction and may be reflected in lower dropout rates as well as rising demand for online courses (Park & Choi, 2009). It has educational value alongside economic efficiency (Appana, 2008;

Bartley & Golek, 2004). In other words, it is important to develop course formats that enable the use of SRL strategies for those who are characterized with a high-conscientiousness personality and openness to experience. However, these course formats should also be suitable to those learners who do not tend to high conscientiousness or openness to experience, and, therefore, may exhibit a lower degree of SRL strategies, or none at all.

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